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Algebra 2 - Tutorial 2

Basic Training Cycle

Vector Spaces

Exercise 1

Are the sets described below vector spaces? Justify your answer without finding a spanning set.

1 $E_1 = \{f \in \mathcal{F}(\mathbb{R}), f(0) = -1\}.$

2 $E_2 = \{(x, y) \in \mathbb{R}^2, xy \in \mathbb{Q}\}.$

3 $E_3 = \{M \in \mathcal{M}_n(\mathbb{R}) \mid \text{all coefficients of } M \text{ are equal}\}.$

4 $E_4 = \{(x, y, z, t) \in \mathbb{R}^4, x + z = y + t \text{ and } y = t^2\}.$

5 $E_5 = \{P \in \mathbb{R}_n[X] \mid P(X + 1) = P(X)\} \text{ where } n \in \mathbb{N}^*.$

6 $E_6 = \{f \in \mathcal{F}(\mathbb{R}), f(x) = 0 \text{ for infinitely many values of } x\}.$

7 $E_7 = \{f \in \mathcal{F}(\mathbb{R}), f \text{ is increasing on } \mathbb{R}\}.$

8 E_8 : The set of nilpotent matrices.

9 $E_9 = \{A \in \mathcal{M}_3(\mathbb{R}) \mid A^t = -A\}.$

10 $E_{10} = \{(u_n)_{n \in \mathbb{N}} \in \mathbb{R}^{\mathbb{N}}, \forall n \in \mathbb{N}, u_{n+2} = u_{n+1} + 4u_n\}.$

11 E_{11} : The set of real arithmetic sequences.

12 $E_{12} = \{(u_n) \in \mathbb{R}^{\mathbb{N}}, u_n u_{n+2} \leq 0\}.$

Exercise 2

Using the notion of *Span*, show that E_3 , E_9 , and E_{11} are vector spaces and determine a basis and the dimension.

Exercise 3

Let F, G, H be vector subspaces of a vector space E .

Show that $F + (G \cap H) \subset (F + G) \cap (F + H)$, and that if $F \subset G$, then the previous inclusion becomes an equality.

Exercise 4

In each of the following cases, determine whether the collection (u_i) generates E :

1 $E = \mathbb{R}^3$ and $u_1 = (1, -1, -2)$, $u_2 = (7, 10, 3)$, $u_3 = (3, -4, -7)$.

2 $E = \mathbb{R}^4$ and $u_1 = (0, 0, 0, 1)$, $u_2 = (0, 0, 1, 1)$, $u_3 = (0, 1, 1, 1)$, $u_4 = (1, 1, 1, 1)$, $u_5 = (1, 1, 1, 0)$.

Exercise 5

Let $u_1 = (1, -2, 1)$ and $u_2 = (0, 1, 2)$.

1 Let $v = (1, 2, 3)$. Is $v \in \text{Vect}(u_1, u_2)$?

2 Determine $k \in \mathbb{R}$ such that $(2k + 1, k, -k) \in \text{Span}\{u_1, u_2\}$.

3 Let $a = (-1, 4, 2)$ and $b = (1, 0, 5)$. Show that $\text{Span}\{u_1, u_2\} = \text{Span}\{a, b\}$.

Exercise 6

For which values of the real number m is the collection $\{u_1, u_2, u_3, u_4\}$ linearly independent in $\mathbb{R}^4$?

$$u_1 = (1, 1, 0, 0), \quad u_2 = (1, m, 1, 0), \quad u_3 = (1, 0, m, 1), \quad u_4 = (1, 0, 0, m)$$

In the case where the sequence of vectors is linearly dependent, provide a dependency relation.

Exercise 7

Let E_1, E_2, E_3 be three vector subspaces satisfying : $E_1 + E_3 = E_2 + E_3$, $E_1 \cap E_3 = E_2 \cap E_3$, and $E_1 \subset E_2$. Prove that $E_1 = E_2$.

Exercise 8

Let F and G be two vector subspaces of $\mathbb{R}^{12}$. Assume that $\dim(F) = 5$, $\dim(G) = 8$, and F is not included in G .

1 Show that $1 \leq \dim(F \cap G)$.

2 Show that $\dim(F \cap G) \leq 4$.

Exercise 9

Let F and G be two subspaces of a vector space of dimension 5 such that $\dim(F) = \dim(G) = 3$. Show that $F \cap G \neq \{0\}$.

Exercise 10

Define $F = \{P \in \mathbb{R}_3[X] \mid P(1) = P(2)\}$.

1 Without computation, show that $\dim(F) \leq 3$.

2 Determine the dimension of F .

Exercise 11

Prove that in each of the following cases, F and G are complementary subspaces of E :

- 1 $E = \{(u_n) \in \mathbb{R}^{\mathbb{N}} \mid \text{convergent sequences}\}$, $F = \{(u_n) \in \mathbb{R}^{\mathbb{N}} \mid \text{sequences converging to } 0\}$ and $G = \{(u_n) \in \mathbb{R}^{\mathbb{N}} \mid \text{constant sequences}\}$.
- 2 $E = \{f \in \mathbb{R}^{\mathbb{R}} \mid f \text{ is differentiable}\}$, $F = \text{Span}(\cosh, \sinh)$ and $G = \{f \in E \mid f(0) = f'(0) = 0\}$.

Exercise 12

We denote by $\mathcal{B}_c = (e_1, e_2, e_3, e_4)$ the standard basis of $\mathbb{R}^4$. Consider the following sets :

$$V = \left\{ \begin{pmatrix} \lambda + \mu \\ 2\lambda + 2\mu \\ -2\lambda + \nu \\ \lambda + 3\mu + \nu \end{pmatrix} \mid (\lambda, \mu, \nu) \in \mathbb{R}^3 \right\}, W = \left\{ \begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix} \in \mathbb{R}^4 \mid \begin{cases} x - y - z + t = 0 \\ 2y + z - t = 0 \\ 3x + y - z + t = 0 \end{cases} \right\}$$

- 1 Justify that V and W are vector subspaces of $\mathbb{R}^4$.
- 2
 - a Show that the vector subspace V has dimension 2 and provide a basis for V . We will denote this basis as (v_1, v_2) .
 - b Show that the vector subspace W has dimension 2 and provide a basis for W . We will denote this basis as (w_3, w_4) .
- 3
 - a Compute the rank of the collection (v_1, v_2, w_3, w_4) .
 - b Is the collection $\{v_1, v_2, w_3, w_4\}$ linearly independent? Does it generate $\mathbb{R}^4$?
 - c Extract a basis of the vector space $F = \text{Span}\{v_1, v_2, w_3, w_4\}$ from the collection $\{v_1, v_2, w_3, w_4\}$.
- 4 Let f_1, f_2, f_3 be the following vectors :

$$f_1 = \begin{pmatrix} 2 \\ 1 \\ -1 \\ 0 \end{pmatrix}, \quad f_2 = \begin{pmatrix} 3 \\ 0 \\ 0 \\ -1 \end{pmatrix}, \quad f_3 = \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix}.$$

- a Show that the collection of vectors $\{f_1, f_2, f_3\}$ is a basis of F .
 - b Show that the collection $\mathcal{B} = \{f_1, f_2, f_3, e_4\}$ is a basis of $\mathbb{R}^4$.
- 5 Determine the coordinate matrix of the vectors of the standard basis $\mathcal{B}_c$ in the basis $\mathcal{B}$.